

Initial Assessment

Recommended Bucketing

I recommend using the MN DOT buckets to allow for easy comparison.

Bucket Name	AADT Range
Sub-7,500	$< 7,500$
Lower-volume evaluation	7,500 – 9,999
Medium-volume evaluation	10,000 – 19,999
High-volume evaluation	$\geq 20,000$

I looked specifically at the 2022 and 2024 conflated data sources separately, looking at the metrics:

This was to treat “Year” as an effective replicate of the analysis, with the added bonus of the road segments contained in the conflation file(s) directly corresponding to interstates of potential interest (the original data Iowa DOT sent, while more numerous, contained estimates on a number of segments not of interest or otherwise beyond the scope of our work).

Initial Results – MN DOT Replication

Summary

Table 2: MnDOT AADT Accuracy Summary — 2022

AADT Bucket	n	MAPE (%)	Median % Error	Mean % Error	APE SD (%)	NRMSE (%)
Sub-7,500	394	47.47	13.45	36.11	150.29	66.25
(7,500-9,999)	93	12.96	4.21	7.34	18.46	22.76
(10,000-19,999)	157	10.98	6.87	5.75	13.76	16.31
(>=20,000)	59	8.59	4.41	4.63	9.13	10.70

Table 3: MnDOT AADT Accuracy Summary — 2024

AADT Bucket	n	MAPE (%)	Median % Error	Mean % Error	APE SD (%)	NRMSE (%)
Sub-7,500	400	38.71	23.99	32.12	46.60	49.14
(7,500-9,999)	92	15.29	8.01	10.18	21.96	25.63
(10,000-19,999)	156	9.57	4.95	4.79	11.94	14.62
(>=20,000)	60	6.79	3.04	4.25	6.87	8.96

MAPE

Table 4: MAPE by AADT Bucket — 2022

AADT Bucket	n	MAPE (%)	MN MAPE (%)
Sub-7,500	394	47.47	NA
(7,500-9,999)	93	12.96	12.16
(10,000-19,999)	157	10.98	6.28
(>=20,000)	59	8.59	5.81

Table 5: MAPE by AADT Bucket — 2024

AADT Bucket	n	MAPE (%)	MN MAPE (%)
Sub-7,500	400	38.71	NA
(7,500-9,999)	92	15.29	12.16
(10,000-19,999)	156	9.57	6.28
(>=20,000)	60	6.79	5.81

Signed Percent Error

Table 6: Signed Percent Error by AADT Bucket — 2022

AADT Bucket	n	Signed % Error	MN Signed % Error
Sub-7,500	394	36.11	NA
(7,500-9,999)	93	7.34	-4.02
(10,000-19,999)	157	5.75	0.26
(>=20,000)	59	4.63	-3.25

Table 7: Signed Percent Error by AADT Bucket — 2024

AADT Bucket	n	Signed % Error	MN Signed % Error
Sub-7,500	400	32.12	NA
(7,500-9,999)	92	10.18	-4.02
(10,000-19,999)	156	4.79	0.26
(>=20,000)	60	4.25	-3.25

Absolute Percent Error Standard Deviation

Table 8: APE Standard Deviation by AADT Bucket — 2022

AADT Bucket	n	APE SD (%)	MN APE SD (%)
Sub-7,500	394	150.29	NA
(7,500-9,999)	93	18.46	8.30
(10,000-19,999)	157	13.76	4.98
(>=20,000)	59	9.13	4.42

Table 9: APE Standard Deviation by AADT Bucket — 2024

AADT Bucket	n	APE SD (%)	MN APE SD (%)
Sub-7,500	400	46.60	NA
(7,500-9,999)	92	21.96	8.30
(10,000-19,999)	156	11.94	4.98
(>=20,000)	60	6.87	4.42

Other

Median Percent Error

Table 10: Median Percent Error by AADT Bucket — 2022

AADT Bucket	n	Median % Error
Sub-7,500	394	13.45
(7,500-9,999)	93	4.21
(10,000-19,999)	157	6.87
(>=20,000)	59	4.41

Table 11: Median Percent Error by AADT Bucket — 2024

AADT Bucket	n	Median % Error
Sub-7,500	400	23.99
(7,500-9,999)	92	8.01
(10,000-19,999)	156	4.95
(>=20,000)	60	3.04

NRMSE

Table 12: NRMSE by AADT Bucket — 2022

AADT Bucket	n	NRMSE (%)
Sub-7,500	394	66.25
(7,500-9,999)	93	22.76
(10,000-19,999)	157	16.31
(>=20,000)	59	10.70

Table 13: NRMSE by AADT Bucket — 2024

AADT Bucket	n	NRMSE (%)
Sub-7,500	400	49.14
(7,500-9,999)	92	25.63
(10,000-19,999)	156	14.62
(>=20,000)	60	8.96